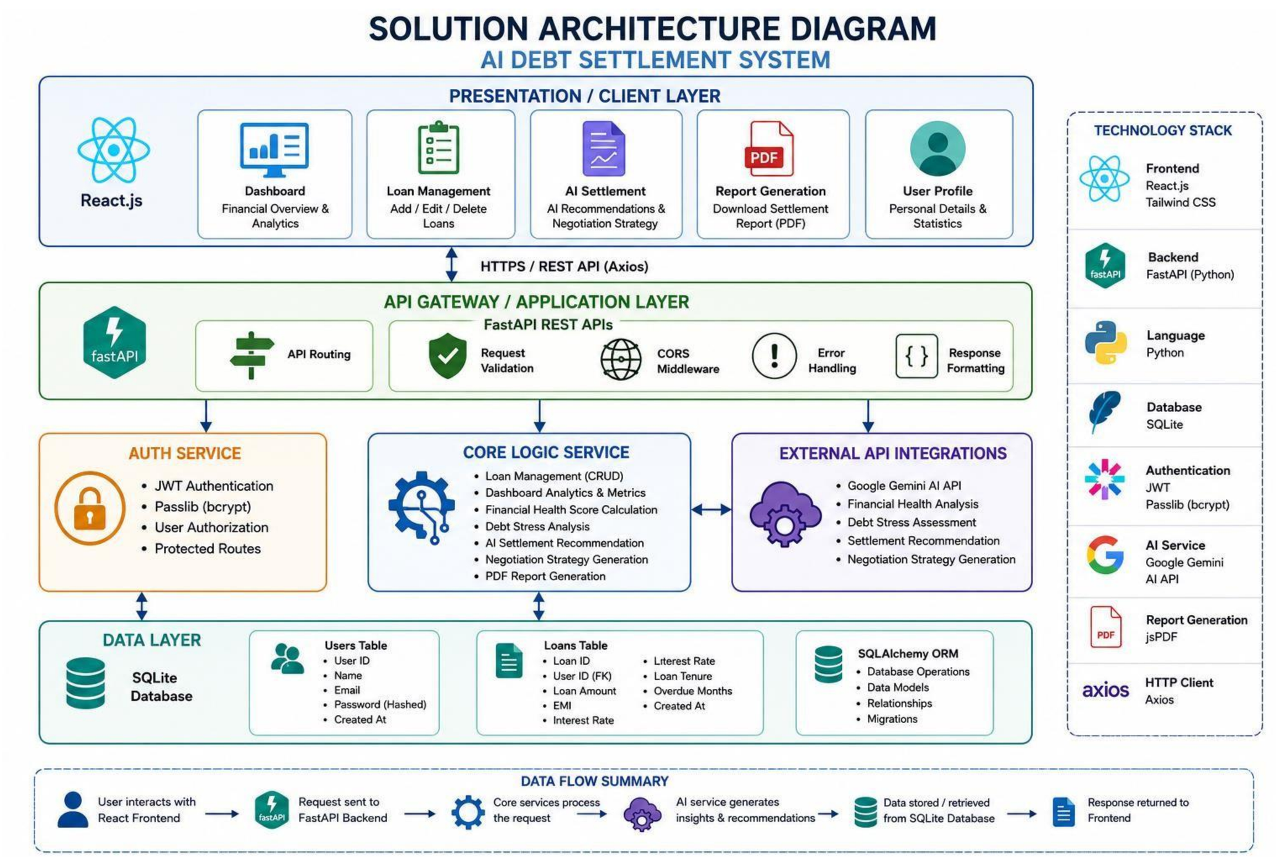
Solution Architecture

Date	15 July 2026
Team ID	xxxxxx
Project Name	AI Debt Settlement System
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface where users can register, log in, manage loans, view dashboards, generate AI settlement reports, download PDFs, and manage profiles. It communicates with the backend through REST APIs.	React.js, HTML5, CSS3, Tailwind CSS, Axios, jsPDF
API Gateway / Application Layer	Receives requests from the frontend, routes them to appropriate backend services, validates requests, handles CORS, processes responses, and exposes REST APIs to the client application.	FastAPI, Python, REST API, Uvicorn
Authentication Service	Authenticates users during login, generates JWT tokens, authorizes protected API access, and securely verifies user credentials.	JWT (JSON Web Token), Passlib (bcrypt), FastAPI Security (HTTPBearer)
Core Logic Service (Microservices)	Implements the application's business logic, including loan management (CRUD), dashboard calculations, financial health analysis, debt stress assessment, AI settlement recommendation generation, negotiation strategies, and PDF report creation.	FastAPI, Python, SQLAlchemy, Google Gemini AI API
External API Integration	Connects the system with Google Gemini AI to generate financial health analysis, settlement recommendations, and negotiation strategies based on loan details.	Google Gemini AI API
Database Layer	Stores and retrieves user accounts, loan records, authentication data, and financial information. Supports persistent data management and query operations.	SQLite, SQLAlchemy ORM
Development & Version Control	Supports application development, debugging, source code management, and collaboration throughout the project lifecycle.	Visual Studio Code, Git, GitHub